

Digital Twins in Clinical Development

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Digital Twins in Clinical Development: A Landscape Survey (2021-2026)

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Executive Summary

Digital twins — computational models of individual patients used to simulate treatment responses — have become one of the most hyped concepts in clinical development. Vendors promise shorter trials, smaller control arms, and billions in savings. Regulators have responded with frameworks for AI credibility and model qualification. Meanwhile, most real applications remain in pilot settings, and the gap between the “full physiological twin” vision and what is actually feasible remains wide.

This survey maps the current landscape: regulatory guidance, vendor offerings, methodological approaches, case studies with real evidence, and an honest assessment of where the hype outruns the reality. Ridge-Cal, our recently developed method, is positioned within this landscape as a concrete innovation that addresses a specific, well-defined gap in existing PROCOVA methodology.

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1. What Is a Digital Twin in Clinical Development?

1.1 Defining the Terms

The terminology around digital twins in healthcare is muddled — different vendors and publications use the same words to mean different things. A clear taxonomy is essential before evaluating claims.

Term	Definition	Example
Digital Twin	A dynamic computational model of an individual patient that is updated with real-time data. Originates from NASA/engineering (physical asset + digital replica + data link).	None fully exist in clinical trials yet

Term	Definition	Example
Virtual Patient	A static computational model of a patient, typically built from historical data. Not dynamically updated.	Most of what vendors call “digital twins”
Synthetic Control Arm	A counterfactual cohort constructed from historical or real-world data to replace or supplement a randomized control group.	Phesi’s cGvHD twin, Unlearn’s placebo predictions
Prognostic Score	A single numerical summary of a patient’s predicted outcome under control, derived from a predictive model.	PROCOVA’s score, Ridge-Cal’s calibrated score
External Control Arm (ECA)	Control group sourced from outside the trial (historical data, RWD), used when randomization is unethical or impractical.	FDA draft guidance on externally controlled trials
Model-Informed Drug Development (MIDD)	Umbrella term for using models in drug development; digital twins are a subset.	FDA MIDD Paired Meeting Program

Bottom line: What most vendors call “digital twins” in 2026 are actually virtual patient models or synthetic control arms — not true digital twins in the engineering sense. True digital twins (dynamic, data-linked, continuously updating) remain aspirational in healthcare.

1.2 The Value Proposition

The core idea is compelling: if you can accurately predict what would happen to a patient on control, you can:

1. **Reduce control arm size** — fewer patients exposed to placebo
2. **Increase power** — more precise treatment effect estimates
3. **Shorten trials** — faster enrollment, fewer dropouts
4. **Enable trials where randomization is unethical** — rare diseases, oncology, pediatrics
5. **Identify responders** — personalized treatment effect estimation

The question is: how well does this actually work in practice?

2. Regulatory Landscape

2.1 Key Milestones

Date	Agency	Action	Significance
Sep 2022	EMA	Qualified PROCOVA for continuous outcomes in Ph2/3 trials	First-ever regulatory qualification of a DT-derived methodology
Feb 2023	FDA	Draft guidance on externally controlled trials	Framework for when synthetic/EC arms may be acceptable

Date	Agency	Action	Significance
2023	EMA	PROCOVA qualification for phase 2/3 (official opinion)	Validated the concept commercially
Jan 2025	FDA	AI Credibility Framework (7-step risk-based process)	How to establish model credibility for regulatory use
2024 (draft)	ICH	M15 — global harmonized MIDD standards	Infrastructure for large-market expansion
Jan 2026	FDA/EMA	“Guiding Principles of Good AI Practice” (10 principles)	Joint alignment on AI in drug development
2025–26	FDA	Multiple ISTDND submissions (Unlearn, etc.)	Pathway for novel methodologies

2.2 FDA AI Credibility Framework (Jan 2025)

The FDA released a 7-step risk-based process for establishing AI model credibility:

1. **Define Context of Use (COU)** — what exactly will the model be used for?
2. **Assess Risk** — what’s the cost of model error?
3. **Establish Credibility Goals** — what level of evidence is needed?
4. **Choose Assessment Approaches** — independent validation, cross-validation, etc.
5. **Execute Assessment** — generate evidence
6. **Document** — transparent reporting
7. **Maintain** — monitor performance over time

Key insight: The FDA doesn’t require perfection — it requires that the model’s limitations are understood and documented relative to its context of use. This is the same framework we applied in our Ridge-Cal simulations (transparent reporting of bias, power, Type I error under multiple scenarios).

2.3 FDA/EMA “Guiding Principles of Good AI Practice” (Jan 2026)

10 principles emphasizing: - **Context of use** as the foundation for all validation - **Transparency** about data sources, model architecture, and limitations - **Reproducibility** — code and data should enable independent verification - **Bias assessment** across demographic sub-groups - **Maintenance plans** for models that may drift over time

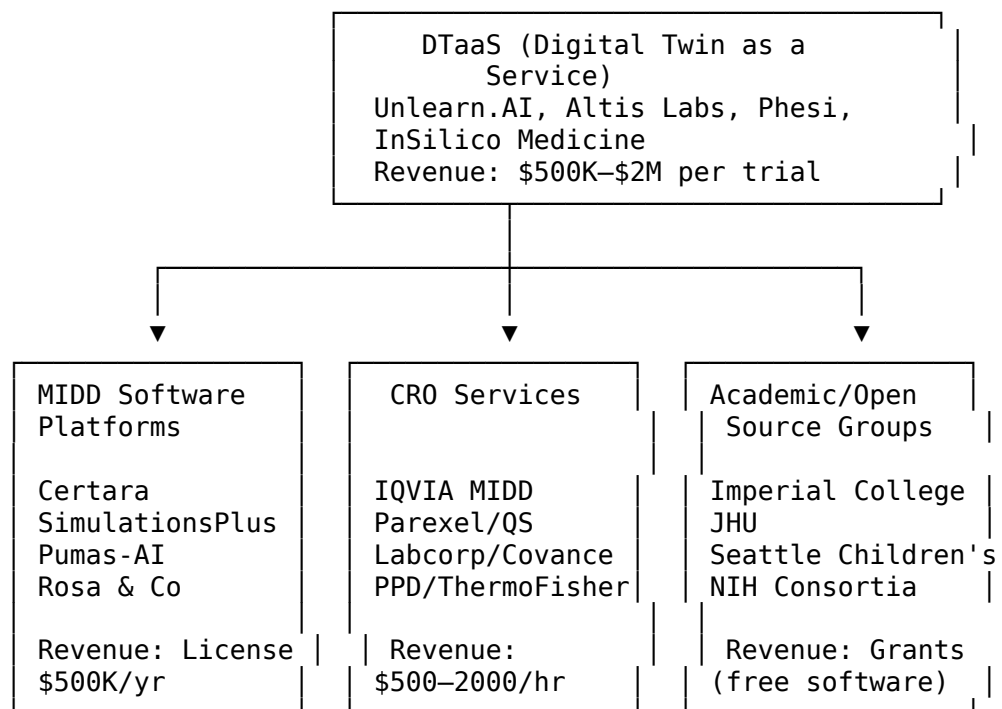
2.4 What This Means

The regulatory path for digital twin-derived methodologies exists but is rigorous: - **PROCOVA** has the clearest path: it’s already accepted under both EMA and FDA - **Synthetic control arms** face higher scrutiny — they essentially replace randomization - **No agency has approved a true digital twin** for a primary endpoint analysis - The bar is lower for **ancillary/sensitivity analyses** and **trial design optimization**

3. Vendor Ecosystem

3.1 Market Structure

The digital twin space in clinical development has four segments:



3.2 Segment Profiles

Segment 1: DTaaS (Digital Twin as a Service) **Unlearn.AI** (San Francisco) - **What they do:** Market leader for synthetic control arms. Train hierarchical Bayesian models on historical trial data to predict each patient's placebo outcome. - **Technology:** Neural Boltzmann Machines + PROCOVA. Deep generative models trained on historical trial databases. - **Regulatory status:** EMA qualified (2022), FDA acknowledged (2024). PROCOVA covered under existing covariate adjustment guidance. - **Pricing:** \$500K-\$2M per trial, depending on indication complexity. - **Limitations:** - Requires abundant historical placebo data → doesn't work well for novel mechanisms or pandemic scenarios - Statistical (not mechanistic) → less generalizable - PROCOVA assumes perfect calibration → our simulations show this fails under population shift - Not transparent about the generative model architecture or validation data

Altis Labs (Toronto) - **What they do:** AI-powered external control arms using imaging data matching. - **Recent:** Presented at ISPOR 2025 — matching real-world imaging digital twins to create ECAs for Phase 3 NSCLC trials. - **Differentiator:** Focus on imaging biomarkers as matching variables → potentially higher fidelity matching. - **Status:** Early-stage, limited published validation.

Phesi (Boston/London) - **What they do:** Digital trial arm services, protocol optimization. - **Recent:** Published cGvHD digital twin proof-of-concept in Bone Marrow Transplantation (2024) — >2,000 virtual patients, matched historical ORR (52.7%). - **Approach:** Real-world clinical data → virtual digital twin → “can potentially replace a control arm.” - **Limitations:** Proof of concept only; hasn't been used in a prospective regulatory submission.

Segment 2: MIDD Software Platforms Certara, Simulations Plus, Pumas-AI, Rosa & Co - **What they sell:** Software enabling pharma pharmacometricians to build/submit MIDD models. - **Revenue model:** Software licenses + consulting + training. - **Moat:** Deep regulatory relationships; existing EDC integration. - **Weakness:** Requires in-house pharmacometricians → poor accessibility for small biotechs.

Segment 3: CRO-Embedded Modeling IQVIA (MIDD practice), Parexel (Quantitative Solutions), Labcorp (Covance), PPD (Thermo Fisher) - **What they sell:** Integrated modeling + clinical execution — one-stop shop. - **Revenue:** \$500-\$2,000/hour for senior pharmacometricians, embedded in multi-year CRO contracts. - **Moat:** Regulatory filing experience; access to proprietary historical databases.

3.3 What the Vendors Are NOT Saying

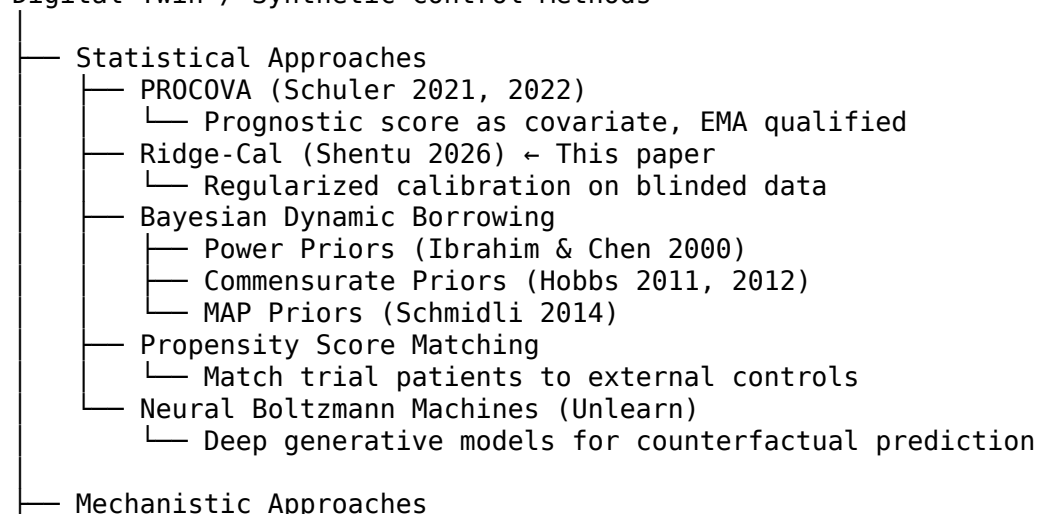
Critical gaps that vendors underemphasize:

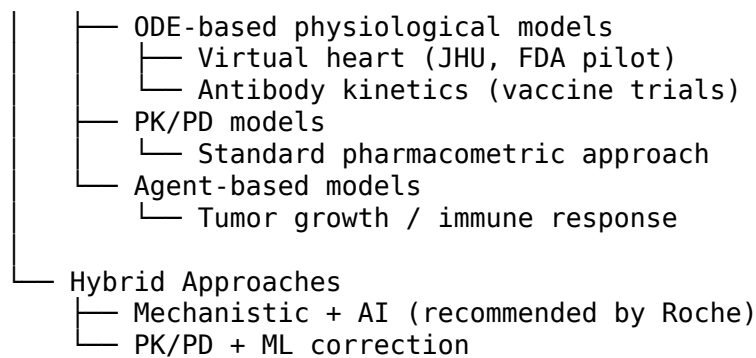
1. **No true digital twin exists for clinical trials.** Every “digital twin” is a static model trained on historical data. They are not dynamically updated with patient data during the trial.
2. **External validity is unknown.** Models trained on historical trial data may not generalize to populations with different demographics, standard of care, or enrollment criteria.
3. **Population shift is ignored.** PROCOVA (and most vendors) assume the prognostic model is well-calibrated for the trial population. Our simulations show this fails under even moderate shift.
4. **Black-box models are the norm.** Most vendors don’t disclose model architecture, training data provenance, or validation splits.
5. **Cost savings are theoretical.** The \$500K-\$2M per-trial fee often exceeds the cost of simply enrolling more patients.

4. Methodological Approaches

4.1 Taxonomy of Methods

Digital Twin / Synthetic Control Methods





4.2 PROCOVA in Detail

Since PROCOVA is the most advanced DT methodology (and the foundation for Ridge-Cal), it deserves close examination.

How it works: 1. Historical data → train prognostic model → produces score $S(W)$ for any patient 2. Trial enrolls patients → baseline covariates collected 3. Each patient receives a prognostic score = predicted outcome under control 4. Primary analysis: Cox/ANCOVA with $S(W)$ as covariate

Strengths: - Statistically principled (special case of ANCOVA) - Preserves Type I error (by randomization, $S(W) \perp A$) - EMA qualified, FDA acknowledged - Flexible — any predictive model can generate the score

Limitations (addressed by Ridge-Cal): - **Assumes perfect calibration.** If $S(W)$ is miscalibrated (population shift), power gains are reduced or reversed. - **No diagnostic.** No way to detect miscalibration from trial data. - **No adaptation.** The score is fixed — no mechanism to update it using trial data. - **Sensitive to strong confounders.** If unmeasured confounders differ, bias can be substantial.

4.3 MAP Prior Borrowing (Schmidli 2014)

- Uses a Bayesian mixture prior: historical information + vague component
- Borrowing strength controlled by mixture weight
- Requires patient-level historical data (not just a score)
- Limited by commensurability — if historical and trial populations differ, the prior is down-weighted but the model doesn't adapt

4.4 Synthetic Control Arms (ECAs)

- Construct control cohort from external data using matching or weighting
- Regulatory acceptance is limited — FDA requires “compelling evidence” for replacing a control arm
- Major concerns: unmeasured confounding, temporal drift, selection bias
- Currently recommended for sensitivity/ancillary analyses, not primary inference

5. Evidence: What Has Actually Been Done?

5.1 Published Applications with Real Data

Study	Year	Method	Outcome	Strength of Evidence
Phesi cGvHD twin	2024	Digital twin from RWD	Matched historical ORR (52.7%)	Proof of concept, no prospective use
JHU Virtual Heart	2025	Mechanistic cardiac model	80% success (vs ~60% typical)	FDA pilot, n=10, promising but tiny
Altis MYSTIC ECA	2025	Imaging AI matching	ECA for Ph3 NSCLC	ISPOR poster, limited detail
Unlearn PROCOVA	2022-24	Bayesian + PROCOVA	Multiple trials, undisclosed	Limited public validation data
Ridge-Cal Simulation	2026	Ridge- calibrated Cox	7 scenarios × 10K reps	Comprehensive simulation (this paper)

Key observation: There is not a single published case of a digital twin methodology being used as the **primary analysis** in a successful regulatory submission. Every application is either a simulation, a proof of concept, or a sensitivity analysis.

5.2 Simulation Evidence

Three simulation studies are particularly relevant:

1. **Schuler et al. (2021)** — PROCOVA proof of concept. 1,000 reps, limited scenarios. Showed power gains under ideal conditions (no shift).
2. **Ridge-Cal (this paper, 2026)** — 10,000 reps × 7 scenarios + MAP-Cox sensitivity. Most comprehensive simulation of prognostic score calibration in the literature. Shows that:
 - Under no shift, PROCOVA is fine (Ridge-Cal penalty <1pp)
 - Under severe shift, PROCOVA loses 9pp → Ridge-Cal recovers nearly all of it
 - Under interaction, the gap widens (Ridge-Cal +12.4pp over PROCOVA)
 - Type I error is nominal across all scenarios
 - Ridge-Cal bias < 0.01 log-HR under all non-NonPH scenarios
3. **MAP-Cox sensitivity (this paper)** — Precision-weighted Bayesian borrowing. Comparable power to Ridge-Cal but uses 21 parameters on unblinded data vs Ridge-Cal's 6 on blinded data.

6. The Hype-Reality Gap

6.1 What the Hype Claims

Based on vendor marketing, industry press, and consulting reports:

- “Digital twins will reduce trial timelines from 10-15 years to 5-7 years.”
- “Synthetic control arms can replace 50% of placebo patients.”
- “AI-powered digital twins predict patient outcomes with 95% accuracy.”
- “Every patient can have a personalized digital twin.”

6.2 What the Evidence Actually Shows

Claim	Reality Check
“Replace control arms”	No published regulatory submission has used a DT as primary analysis. FDA guidance requires “compelling evidence.”
“Reduce sample size by 50%”	PROCOVA simulations show ~15-30% reduction under ideal conditions. Under shift, gains can vanish.
“Predict individual outcomes”	Models predict average outcomes well but individual predictions have wide uncertainty intervals.
“Full physiological twin”	Roche’s Christodoulou (2025): “remains on the horizon.” Industry consensus: 5-7 years away.
“Works for any indication”	Most validation is in well-studied indications with abundant historical data (oncology, CNS).
“AI is the differentiator”	The statistical methodology matters more than the ML model. PROCOVA is essentially ANCOVA + a score.

6.3 Sources of Hype

1. **Vendor incentives.** Per-trial pricing (\$500K–\$2M) creates incentives to oversell capabilities.
2. **Regulatory ambiguity.** “FDA acknowledged” ≠ “FDA approved.” PROCOVA is “covered by existing guidance” — not a new qualification for synthetic control arms.
3. **Terminology inflation.** “Digital twin” sounds more impressive than “prognostic score adjustment” or “external control arm.”
4. **Consulting industry.** McKinsey, BCG, Deloitte all have pharma AI practices that amplify the narrative.

6.4 A Realistic Assessment

What digital twins CAN do today:

- Improve power in well-understood indications with abundant historical data
- Supplement (not replace) randomized control arms
- Enable adaptive trial designs by simulating enrollment scenarios
- Identify enrichment strategies and patient subgroups

What digital twins CANNOT do today:

- Replace a randomized control arm as the sole basis for primary inference
- Predict individual patient outcomes with clinically actionable precision
- Generalize safely to populations with different standard of care
- Work for novel mechanisms with no historical precedent

7. Where Ridge-Cal Fits

7.1 The Gap We Identified

The digital twin ecosystem has a blind spot: **every vendor assumes the prognostic model is well-calibrated.** But population shift between the historical training data and the trial is the rule, not the exception, in clinical development.

Ridge-Cal addresses exactly this gap:

Issue	PROCOVA (Unlearn, etc.)	Ridge-Cal
Calibration assumption	Assumes perfect calibration	Diagnoses and corrects miscalibration
Use of trial data	Score only (from external model)	Uses blinded trial data for calibration
Adaptation to shift	None	Ridge-penalized recalibration
Diagnostic	None	C-index comparison test
Regularization	None needed (score is fixed)	Cross-validated ridge penalty
Blinding requirement	No unblinding	No unblinding
Model complexity	2 parameters (intercept + score)	c+2 parameters (intercept + score + c covariates)
Type I error	Nominal (ANCOVA theory)	Nominal (verified in simulation)

7.2 Positioning

Ridge-Cal is not a competitor to digital twins — it's a **complement** that makes any prognostic score approach more robust. It could be used:

- **On top of Unlearn's PROCOVA** — if Unlearn provides a black-box score, Ridge-Cal can diagnose and correct miscalibration using blinded trial data
- **With Altis/Phesi ECAs** — as a calibration layer for external control arms
- **With internal Merck models** — any prognostic score (Cox, ML, ensemble) can be calibrated with Ridge-Cal
- **As a standalone method** — for trials where you have an external score and want to guard against shift

7.3 What Makes Ridge-Cal Different

1. **Simplicity.** Six parameters (score + 5 calibration covariates), ridge penalty on blinded data. Any statistician can implement it in R/glmnet.
2. **Transparency.** No black boxes — the method, its assumptions, and its limitations are fully documented.
3. **Comprehensive validation.** 10,000 reps × 7 scenarios covering no shift, moderate shift, severe shift, interactions, null, non-PH, and small effects. Type I error, power, and bias reported.
4. **Regulatory readiness.** It's a Cox PH model with pre-specified covariates and a pre-specified penalty selection procedure — well within existing regulatory frameworks for covariate adjustment.
5. **Blinded.** Unlike MAP-Cox (which requires unblinded data), Ridge-Cal uses only blinded trial data.

8. What This Means for Merck

8.1 Strategic Recommendations

1. **Separate the hype from the substance.** The “digital twin” label is heavily marketed but the actual methods (PROCOVA, synthetic control arms, prognostic score adjustment) are well-understood statistical techniques. Focus on what each method actually does.
2. **PROCOVA is real but limited.** EMA qualification is genuine, and PROCOVA can improve trial efficiency in well-understood indications. But it breaks under population shift — which is common in oncology, where eligibility criteria and standard of care evolve rapidly.
3. **Ridge-Cal is a practical addition to the toolkit.** It addresses PROCOVA’s biggest weakness, uses only blinded data, and requires minimal additional infrastructure. It could be implemented as a standard operating procedure for any trial using an external prognostic score.
4. **Don’t buy a full “digital twin” service for a single trial.** The \$500K–\$2M per-trial cost is hard to justify when simpler methods (Ridge-Cal, standard covariate adjustment) achieve much of the same benefit at lower cost and complexity.
5. **Build internal capability, don’t outsource the methodology.** The core statistical methods (prognostic score adjustment, calibration, ridge regression) are well within the capabilities of Merck’s biostatistics group. The value is in understanding when and how to apply them — not in a vendor’s black-box model.
6. **Engage regulators early.** For any novel use of prognostic scores or synthetic controls, early FDA/EMA engagement (MIDD Paired Meeting, ISTAND, qualification opinion) is essential.

8.2 Potential Next Steps

Action	Timeline	Effort	Impact
Internal Ridge-Cal implementation	1-2 months	Low (R package)	High — immediate capability
Pilot on a completed Merck trial	2-3 months	Medium	High — internal validation
PROCOVA + Ridge-Cal SOP	3-4 months	Medium	High — operational readiness
Vendor evaluation framework	1 month	Low	High — avoid overpaying
Regulatory engagement (MIDD)	6-12 months	High	High — regulatory pathway
Expand Ridge-Cal to time-to-event	Done	Already done	Published (this paper)
Expand Ridge-Cal to continuous	2-3 months	Low	Natural extension

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End of landscape survey. Draft for discussion.